



Hallucination Mitigation in Reasoning Models

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Introduction

Focus: Story Continuation with Hallucination Control

*Given a 4-sentence context, the model generates a single ending sentence.
The dataset also provides one reference ending for comparison and analysis.*

The key **challenge** is not merely creativity, but semantic consistency and commonsense plausibility: whether the generated ending remains faithful to the established context.

Our **goal** is to fine-tune a smaller-size model (building on a strong prompt baseline and preference signals) to reduce hallucinations in this semantic reasoning-oriented generation setting, improving coherence and overall narrative validity.



Why is this interesting and challenging?

Specific Goals

The model must produce an ending that is:

- coherent with prior events, plausible and commonsense-consistent
- stylistically natural as a short narrative closing

Failure Modes

- abrupt topic shifts
- contradictions
- generic or dull endings
- invented details that break the story logic



Related Work

1. **Hallucination in Large Language Models** (Huang et al. (2025)): factual and faithfulness hallucinations - attributed to misinformation, bias, knowledge boundary problems, and limitations in training stages
2. **Process and Outcome Reward Models** (Xie et al. (2025)): Introduces SP-PRM constructs a process reward model (PRM) from an ORM using partial-sequence data, enforcing score and preference consistency.
3. **System 1 to System 2 Reasoning** (Zhou et al. (2025)): Surveys the transition from fast heuristic "System-1" behavior to deliberative "System-2" reasoning in LLMs.
4. **Why Language Models Hallucinate** (Kalai et al.): the pretraining objective itself, not just flawed data, leads models to generate errors.
5. **Reasoning Models Hallucinate More** (Li and Ng): reinforcement learning-based fine-tuning for reasoning, while improving benchmark performance, increases hallucinations



Data

ROCStories: 5-sentence short stories (4 context sentences + ending)

Evaluation: random sample of 500

Reward model training (3000 judged samples):

- sample 1,000 prompts from local model
- generate $2(2 \times 1000)$ candidate endings perx prompt from gpt

DPO Training: 1,000 GPT-4-generated high-quality continuations with local low-quality to construct preference pairs for DPO training

Split	Stories	Purpose
Test Set	500	Evaluation
RM Training	3,000	Reward Model Training
DPO Training	1,000 pairs	Preference Alignment

Table 1: Dataset Statistics and Usage.



Evaluation

LLM-as-a-Judge + GPT-4.1-mini with our rubric.

Scoring 0-5 on the following dimensions:

1. Information Completeness
2. Factual Accuracy
3. Relevance
4. Logical Coherence
5. Creativity

We also introduce a Quality Gated Length Adjustment, P_{verb}

$$S_{quality} = S_{raw} + \lambda \cdot B_{len} - P_{verb}$$

Where:

- S_{raw} is the unweighted mean of the five dimension scores.
- $B_{len} \in [0, 1]$ is a bonus term that rewards outputs within the target range of 40–120 words.
- $\lambda = \left(\frac{IC+REL}{10}\right)^2$ is a quality gate. This ensures that only answers with high Information Completeness and Relevance benefit from the length bonus.
- P_{verb} is a penalty applied if the length exceeds 120 words without sufficient coherence.



Code Demo



Baseline + Extension 1

Loose: Minimal guidance requesting only a story continuation.

Score = 2.955

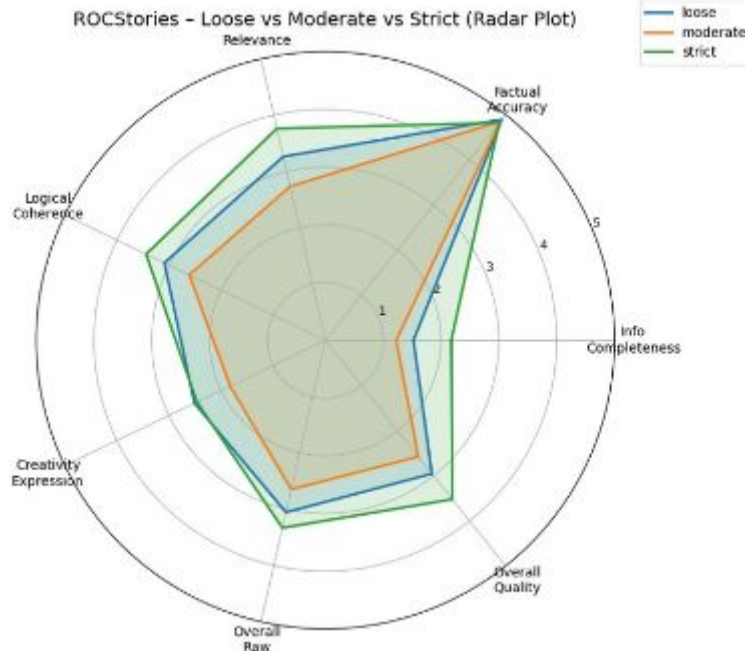
Moderate: Includes reasoning guidance with flexible structure.

Score = 2.577

Strict (Ext 1): Enforces a four-step template.

Score = 3.533

Strict template shows substantial gains in Relevance (+0.50) and Logical Coherence (+0.35) while maintaining high Factual Accuracy





Extension 2: Reward-based reranking

Uses a Reward Model (RM) for best-of-N inference-time reranking to approximate GPT-4.1-mini's rubric-based evaluation.

Training: 3,000 samples (2,000 GPT-4 continuations + 1,000 lower-scoring DeepSeek outputs), labeled with 5 rubric dimensions.

Inference: Generate N=3 strict-prompt continuations; select highest RM score.

Results: Overall Quality = 3.503 (-0.019 vs. strict baseline).

Improves Relevance (+0.21) but reduces Creativity (-0.37), indicating safer output



Extension 3: DPO preference alignment

Directly optimize the policy from pairwise preferences, avoiding inference-time reranking or sampling.

Training: 1,000 preference pairs (x, y+, y-) from GPT-4.1-mini judgments. LoRA-based DPO fine-tuning on the base DeepSeek model; no reward model needed at inference.

Inference: Generate N=3 strict-prompt continuations; select highest RM score.

Results: Overall Quality = 3.961 (+12.5% vs. strict CoT baseline). Improves all dimensions: Information Completeness (+0.48), Relevance (+0.45), Logical Coherence (+0.47), Creativity (+0.66)

Unlike RM reranking, DPO learns to generate higher-quality and more creative outputs



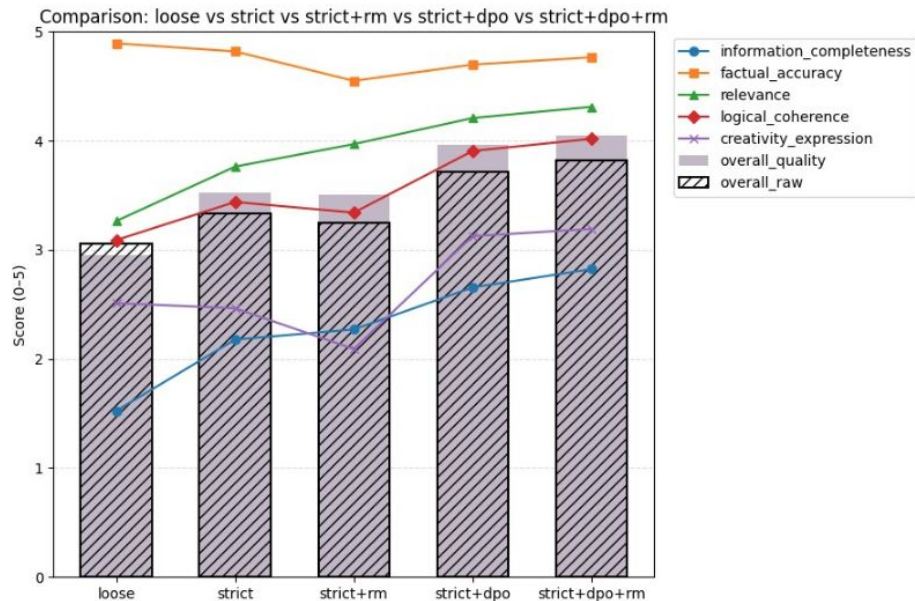
Combine: DPO + Reward

Combine training-time alignment (DPO) with inference-time selection (RM).

Method: DPO-trained policy generates N=3 candidates; RM selects the highest-scoring continuation.

Inference: Generate N=3 strict-prompt continuations; select highest RM score.

Results: 4.044 (best overall)
+36.8% vs. simple baseline,
+14.8% vs. strict CoT baseline.





Results

Baseline: 3.52 overall quality

RM: 3.50 overall quality, with higher relevance (+0.21) but lower creativity (-0.37)

DPO: 3.96 overall quality, with higher information completeness (+0.48), relevance (+0.45), logical coherence (+0.47), and creativity (+0.66)

RM + DPO: 4.04 overall quality

Decoding settings (temperature/top-p/top-k) are kept consistent across comparisons

Only policy (base vs DPO) and reranking (on/off) and N are varied.

Method	IC	FA	REL	LC	CE	OQ
Loose Baseline	1.524	4.893	3.264	3.090	2.511	2.955
Moderate CoT	1.226	4.844	2.734	2.611	1.826	2.577
Strict CoT	2.177	4.819	3.763	3.439	2.464	3.522
Strict + RM	2.270	4.549	3.969	3.341	2.091	3.503
Strict + DPO	2.655	4.698	4.208	3.906	3.125	3.961
DPO + RM	2.823	4.765	4.311	4.018	3.192	4.044



Analysis

The Baseline Problem: "Prompt Adherence Paradox"

- Loose Prompt: High Variance → Creative but Hallucinates.
- Strict Prompt: Rigid Adherence → Safe but Repetitive.

The Partial Solution: Reward Model (Inference-Time)

- Action: Filters out low-quality/hallucinated answers.
- Result: High Factual Accuracy, but Creativity Collapsed (-0.37).
- Verdict: The model learned that "boring is safe."

The Full Solution: DPO (Training-Time)

- Action: Reshapes the underlying probability distribution.
- Result: Creativity Restored (+0.66) while maintaining safety.
- Verdict: Alignment teaches the model to be both accurate and engaging.



Error Analysis / Limitations

The "Prompt Adherence Paradox"

- Observation: Moderate prompting performed worse than Loose prompting (2.58 vs. 2.96).
- Cause: "Instruction Confusion"—the model outputted planning steps instead of the narrative.
- Takeaway: Reasoning constraints must be binary (loose or strict); "guiding" instructions confuse smaller models.

DPO Failure Mode: "Silence as Safety"

- Issue: Initial DPO policy resulted in a 23.2% refusal rate (empty outputs).
- Mechanism: The model learned that abstaining was statistically safer than risking a hallucination penalty.
- Mitigation: Enforced min_new_tokens=32 to overcome the initial activation barrier.

Methodological Constraints

- Creativity vs. Safety: RM reranking alone reduced Creativity (-0.37) by favoring generic, "safe" answers.
- Model Scale: Experiments limited to 8B parameters (DeepSeek-R1-Distill); larger models may resist the "Prompt Paradox."
- Judge Reliability: GPT-4o-mini judge may favor verbose outputs or miss subtle logical contradictions.



Final Takeaways

Structured Prompting works.

Step-by-step structure → +19% quality vs. baseline.

Reward Models help selection.

Improve relevance and reduce hallucinations, but can limit creativity.

DPO provides the largest single improvement.

Largest gain: better one-shot quality and restored creativity.

DPO + RM are complementary.

Combined pipeline → ~37% improvement overall.

The combination of RM and DPO offers a total improvement of 36.8%!